SOP-BUF-002 Preparation of Salt Stock Solutions (NaCl, KCl, MgCl2)

DEMO DATA - written for the SmartLab AI demonstration. This is not an approved laboratory procedure.

1. Scope
This procedure covers sodium chloride, potassium chloride and magnesium chloride hexahydrate stock solutions prepared by weight.

2. Safety
Wear safety glasses, a lab coat and gloves. Clean up spilled salt promptly; wet floors are slippery.

3. Procedure
Calculate the mass of salt from its molecular weight, the target concentration and the final volume. Weigh the salt and dissolve it in deionised water equal to about 80% of the final volume. Magnesium chloride hexahydrate is hygroscopic: weigh it quickly and keep the container closed. Make up to the final volume with deionised water and mix by inversion. Sterilise by autoclaving at 121 C for 20 minutes or filter through a 0.22 um membrane.

4. Storage
Store sodium chloride and potassium chloride stocks at room temperature for up to 6 months. Store magnesium chloride solutions at 4 C.